Aveon Client Installation Manual

1. Purpose

This manual explains how to install and run the Aveon Biometric Device Management software on a client Windows machine.

2. System Requirements

- OS: Windows 10 / Windows 11 / Windows Server
- RAM: 8 GB minimum (16 GB recommended)
- CPU: Intel i5 or above (or equivalent)
- Disk: 20 GB free space minimum
- Network: LAN access to biometric devices

Required Software

- Python 3.9+ (added to PATH)
- Node.js 18+ (includes npm)
- Microsoft SQL Server (Express/Standard/Enterprise)
- ODBC Driver 18 for SQL Server
- Biometric SDK installed and COM registered (zkemkeeper) on the server machine

3. Package Contents

Deployment package includes:

- backend/
- frontend/
- database/schema.sql
- install-and-start.cmd (single-click installer)
- install-and-start.ps1
- scripts/install-client.ps1
- scripts/start-client.ps1
- scripts/register-services.ps1

4. Database Setup

1. Open SQL Server Management Studio (SSMS).
2. Create a new database (example: AIFace).
3. Open and execute:
- database/schema.sql
4. Confirm tables are created and admin user is inserted.

5. Configuration

5.1 Backend Environment

Edit backend/.env:

```
APP_ENV=production
APP_DEBUG=false
SECRET_KEY=replace_with_strong_secret
API_V1_PREFIX=/api/v1

ADMIN_USERNAME=admin
ADMIN_PASSWORD=admin123

SQLSERVER_SERVER=localhost
SQLSERVER_DATABASE=AIFace
SQLSERVER_USER=sa
SQLSERVER_PASSWORD=your_sql_password
SQLSERVER_DRIVER=ODBC Driver 18 for SQL Server
SQLSERVER_ENCRYPT=no

CORS_ORIGINS=http://localhost:5173,http://127.0.0.1:5173
WEBHOOK_URL=
```

5.2 Frontend Environment

Edit frontend/.env:

```
VITE_API_BASE_URL=http://localhost:8000/api/v1
VITE_BRAND_LOGO_URL=/branding/logo.png
```

To use custom logo:

1. Copy logo image to: frontend/public/branding/logo.png
2. Or set another path in VITE_BRAND_LOGO_URL.

6. Installation Methods

Method A: Single-Click Installation (Recommended)

1. Right-click install-and-start.cmd
2. Select **Run as Administrator**

This automatically:

- installs backend dependencies
- installs frontend dependencies
- builds frontend
- starts backend (port 8000) and frontend (port 5173)

Method B: Manual Installation

From package root:

```
.\scripts\install-client.ps1
.\scripts\start-client.ps1
```

7. Access URLs

- Frontend: http://localhost:5173
- Backend health check: http://localhost:8000/health

Default login:

- Username: admin
- Password: admin123

Change admin password after first login.

8. Service Mode (Auto-start after Reboot)

Use NSSM (Non-Sucking Service Manager).

1. Install NSSM.
2. Run:

```
.\scripts\register-services.ps1
```

This creates and starts:

- AveonBackend
- AveonFrontend

9. Firewall and Network

Allow inbound ports if remote users access the app:

- 5173 (frontend)
- 8000 (backend)

If same machine only, localhost access is enough.

10. Device Connection Checklist

For each device:

1. Confirm device and server are on same network.
2. Set correct server URL on device (iClock mode if used):
• http://<server-ip>:8000/iclock
3. In app, add device with correct IP address.
4. Use **Test SDK** and **Connect**.
5. Verify device users can be loaded.

11. Troubleshooting

Backend not starting

- Check Python installation and PATH
- Verify SQL credentials in backend/.env
- Ensure ODBC Driver 18 is installed

Frontend blank page

- Confirm VITE_API_BASE_URL is correct
- Rebuild frontend: npm run build

Device actions fail

- Check SDK installed and zkemkeeper COM registration
- Confirm device IP/port (default 4370)
- Ensure no firewall blocks device communication

Login fails

- Verify backend is running
- Check admin credentials in DB and .env

12. Post-Installation Handover

After successful installation:

- Share application URL with client
- Provide user accounts and passwords
- Run basic validation:
- Add device
- Connect device
- Load users
- Enroll test user
- Verify attendance log

13. Backup and Maintenance

- Take daily SQL backup
- Keep package version and changelog
- Rotate admin password periodically
- Apply updates using latest deployment package